

California House Price Prediction using Machine Learning

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Role: Aspiring AI & Machine Learning Engineer

1. Project Overview

This project presents an end-to-end supervised machine learning solution for predicting California house prices. It covers data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, hyperparameter tuning, model serialization, and deployment through a Streamlit web application.

2. Objective

To build an accurate regression model capable of estimating house prices from housing characteristics while comparing multiple supervised learning algorithms to identify the best-performing model.

3. Dataset

The California Housing dataset was used. Features include longitude, latitude, housing median age, total rooms, total bedrooms, population, households, median income, and ocean proximity. The target variable is median_house_value.

4. Methodology

The workflow includes data loading, preprocessing, missing value analysis, EDA, feature engineering, label encoding, feature scaling, train-test split, model training, hyperparameter tuning, model evaluation (R^2 , MAE, MSE, RMSE), model saving using Pickle, and deployment using Streamlit.

5. Models Used

Linear Regression, Random Forest Regressor, and XGBoost Regressor were implemented and compared to select the best-performing model.

6. Deployment

The trained model, scaler, and encoder were integrated into a Streamlit application that accepts user inputs, preprocesses them, and predicts California house prices.

7. Technologies Used

Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Streamlit, and Pickle.

8. Conclusion

This project demonstrates the complete machine learning lifecycle from raw data to deployment and showcases practical skills in supervised learning, feature engineering, model evaluation, and web application development.

9. Future Scope

Future enhancements include advanced ensemble techniques, explainable AI (SHAP), cloud deployment, CI/CD pipelines, and automated retraining.

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